Linear Algebra on a Computer

An Introduction to Black Box Methods

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Outline

Symbolic Computation

Superset of computer algebra

- Symbols or exact arithmetic
- Exact finite representation of mathematical structures
- Abstract structures (groups, rings, fields, etc.)
- Polynomials and power series
- Linear algebra
- Number theory
- Algebraic geometry
- Computational group theory
- Differential equations
- Automated theorem proving

Computer Algebra Systems

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE, SAGE

Special Purpose Systems

- CoCoA (Computations in Commutative Algebra)
- GAP (Groups, Algorithms, and Programming)
- NTL (Number Theory Library)
- SINGULAR (polynomial computations)
- *Theorema* (automated theorem proving)

Long History

Ancient Algorithms

- Euclidean Algorithm
- Chinese Remainder Algorithm

Isaac Newton's *The Universal Arithmetic* (1728)

Systematically discusses rules for manipulating universal mathematical expressions, that is, formulae containing symbolic indeterminates, and algorithms for solving equations built with these expressions.

Symbolic Computation vs. Numerical Analysis

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly
- Error propagation important (condition number, stability)

Symbolic Computation

- Find exact solution quickly
- May never approximate (Structure may not have a metric)

Algorithms may not be compatible

- Numerical algorithms may never find exact solution
- Symbolic algorithms may be ill-conditioned or unstable

Infinite Mathematical Structure

General Approach

- Computer has finite memory
- Cannot compute exactly over reals, rationals, integers, etc.
- Compute bound M on desired solution
- Solve via modular algorithms & reconstruct solution
  - Chinese Remainder Algorithm
  - Hensel Lifting
  - Rational Number Reconstruction

Selecting the Modulus

Big Prime Method: $m = p$

Small Prime Method: $m = \prod_i p_i$

Small Prime Power Method: $m = p^\ell$

Matrix Determinant

Example

$$A = \begin{bmatrix} -4 & 3 \\ 3 & 1 \end{bmatrix}$$

Hadamard's bound

- $B = \max_{i,j} |a_{i,j}| = 4$
- $|\det A| \leq n^{n/2} B^n = 2^{2/2} 4^2 = 32$

Small Prime Method

- $m > 2 \cdot 32 \implies m = 7 \cdot 11$
- $|\det A| \equiv 1 \pmod{7}$
- $|\det A| \equiv 9 \pmod{11}$
- CRA: $|\det A| \equiv 64 \pmod{77}$
- $\det A = -13$

Probabilistic Algorithms

Monte Carlo Algorithm

- Always fast; Probably correct

Las Vegas Algorithm

- Probably fast; Always correct

Bounded-error, Probabilistic, Polynomial time (BPP)

- Probably fast; Probably correct
- Atlantic City?

Probabilistic Algorithms

Schwartz-Zippel Lemma

Let R be an integral domain, $n \in \mathbb{N}$, $S \subseteq R$ finite with $s = |S|$ elements, and $f \in R[\lambda_1, \dots, \lambda_n]$ a polynomial of total degree at most $d \in \mathbb{N}$.

1. If f is not the zero polynomial, then f has at most ds^{n-1} zeros in S^n.
2. If $s > d$ and f vanishes on S^n, then $f = 0$.

Converting Algorithms

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
    - Solution guaranteed to be correct
    - Return correct answer
- Will not stop early with probability $1 - p$
    - Halt algorithm
    - Return some answer
    - Need not be correct

Problems of Interest in Symbolic Linear Algebra

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

- $\det(A)$ where $A \in \mathbb{F}^{n \times n}$
- A nonsingular

Matrix Rank

- $\operatorname{rank}(A)$ where $A \in \mathbb{F}^{n \times n}$
- A singular

Solving Linear Systems

Example (Nonsingular system)

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 3 & 3 & -2 & 3 \\ 0 & -3 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix}$$

Gaussian Elimination

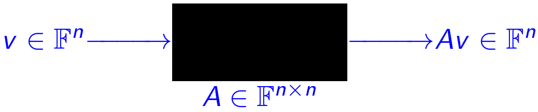
Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

- Row echelon form and back substitution
- Must know how matrix stored
- Matrix changed in calculation
- Sparse matrices may become dense

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix
- Only matrix-vector products allowed
- Independent of implementation
- Implementations may be efficient in time or space
- Not unique to symbolic computation
- Can compute Krylov sequence $(A^i v)_{i \in \mathbb{Z}_{\geq 0}}$

Possible Black Box Matrices

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Storage: n^2 (store every entry)

Time: $O(n^2)$

Sparse Matrix

Only η nonzero entries

Storage: $O(\eta)$ (store nonzero entries and indices)

Time: $O(\eta)$

Possible Black Box Matrices

Hilbert Matrix

$$A = \begin{bmatrix} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots & \frac{1}{n+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{2n} \end{bmatrix}$$

Storage: O(1) (store only dimension n)

Time: O(n)

Possible Black Box Matrices

Toeplitz Matrix

$$A = \begin{bmatrix} a_n & a_{n+1} & a_{n+2} & \cdots & a_{2n-1} \\ a_{n-1} & a_n & a_{n+1} & \cdots & a_{2n-2} \\ a_{n-2} & a_{n-1} & a_n & \cdots & a_{2n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_n \end{bmatrix}$$

Storage: $\mathrm{O}(n)$

Time: $\mathrm{O}(n \log(n) \log\log(n))$ (implement via fast polynomial multiplication)

Linearly Recurrent Sequences

Definition

- Let $\mathbb{V}$ be a vector space over field $\mathbb{F}$.
- A sequence

$$a = (a_i)_{i \in \mathbb{Z}_{\geq 0}} \in \mathbb{V}^{\mathbb{Z}_{\geq 0}}$$

is *linearly recurrent* if and only if there exist $m \in \mathbb{Z}_{\geq 0}$ and

$$c_0, \ldots, c_m \in \mathbb{F}, \quad c_m \neq 0$$

such that for all $i \geq 0$

$$\sum_{j=0}^{m} c_j \; a_{i+j} = 0$$

$$a_{i+m} = -\frac{1}{c_m} \left(\sum_{j=0}^{m-1} c_j \; a_{i+j} \right)$$

Linearly Recurrent Sequences

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $\mathbb{V} = \mathbb{F} = \mathbb{R}$
- $a_{i+2} = a_{i+1} + a_i$
- $a_{i+2} - a_{i+1} - a_i = 0$

Generating Polynomials

Definition

The polynomial $f(\lambda) = \sum_{j=0}^{m} c_j \, \lambda^j$ generates a

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $a_{i+2} - a_{i+1} - a_i = 0$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a

Generating Polynomials

Module

- Define $f \bullet a = \left(\sum_{j=0}^{m} c_j \; a_{i+j} \right)_{i \in \mathbb{Z}_{\geq 0}} = (0)_{i \in \mathbb{Z}_{\geq 0}} = 0 \in \mathbb{V}^{\mathbb{Z}_{\geq 0}}$
- $\mathbb{V}^{\mathbb{Z}_{\geq 0}}$ is an $\mathbb{F}[\lambda]$-module
- If $f \bullet a = 0$ and $g \in \mathbb{F}[\lambda]$, then

$$(g \; f) \bullet a = g \bullet (f \bullet a) = g \bullet 0 = 0$$

Generating Polynomials

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a
  - $a_{i+3} - 2a_{i+1} - a_i = 0$
  - $a_{i+3} = 2a_{i+1} + a_i$
- $\lambda^k f = \lambda^{k+2} - \lambda^{k+1} - \lambda^k$ also generates a
  - $a_{i+k+2} - a_{i+k+1} - a_{i+k} = 0$
  - $a_{i+k+2} = a_{i+k+1} + a_{i+k}$
  - Skips first k elements of a

Minimal Generating Polynomial

Existence

- $\{f \in \mathbb{F}[\lambda] \mid f \bullet a = 0\}$ is an ideal
- $\mathbb{F}[\lambda]$ is a principal ideal domain
- There exists a unique monic generator of minimal degree, the *minimal generating polynomial* of sequence
- Minimal polynomial divides all generating polynomials

Example (Fibonacci Numbers)

The polynomial $f = \lambda^2 - \lambda - 1$ is the minimal generating polynomial of the Fibonacci sequence a.

Matrix Power Sequence

Matrix Power Sequence

- $f \bullet (A^i)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) = 0$
- $\det(\lambda I - A)$ generates $(A^i)_{i \in \mathbb{Z}_{\geq 0}}$
  - Cayley-Hamilton Theorem
- Let f^A be the minimal polynomial of $(A^i)_{i \in \mathbb{Z}_{\geq 0}}$.
  - $f^A \mid \det(\lambda I - A)$
  - $\deg(f^A) \leq \deg(\det(\lambda I - A)) = n$

Krylov Sequence

Krylov Sequence

- $f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A)\ v = 0$
- $f(A) = 0$ means $f(A)\ v = 0$
- $\{f \mid f \bullet (A^i)_{i \in \mathbb{Z}_{\geq 0}} = 0\} \subseteq \{f \mid f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}} = 0\}$
- Let $f^{A,v}$ be the minimal polynomial of $(A^i v)_{i \in \mathbb{Z}_{\geq 0}}$.
    - $f^{A,v} \mid f^A \mid \det(\lambda I - A)$
    - $\deg(f^{A,v}) \leq \deg(f^A) \leq \deg(\det(\lambda I - A)) = n$

Solving a Linear System

Using Linear Recurrences to Solve a Linear System

- Suppose $g = \sum_{j=0}^{m} c_j \; \lambda^j$, $g(0) = c_0 \neq 0$, and $g \bullet (A^i b)_{i \in \mathbb{Z}_{\geq 0}}$
- If g exists, then $f^{A,b}$ satisfies the requirements.
- Consider linear combination $c_0 \; b + c_1 \; A \; b + \cdots + c_m \; A^m \; b = 0$
- $b = -\frac{1}{c_0} \sum_{j=1}^{m} (A^j \; b) = A \left(-\frac{1}{c_0} \sum_{j=1}^{m} (c_j \; A^{j-1} \; b) \right)$
- $x = -\frac{1}{c_0} \sum_{j=1}^{m} (c_j \; A^{j-1} \; b)$ is a solution

Nonsingular Matrix

Nonsingular A

- $\det(A) \neq 0$
- $\det(\lambda I - A)|_{\lambda=0} \neq 0$
- $f^{A,b}(0) \neq 0$
- $x = -\frac{1}{c_0} \sum_{j=1}^{m} (c_j \; A^{j-1} \; b)$ is the unique solution

Computing the Minimal Polynomial of Krylov Sequence

Bilinear Projection Sequence

- $f \bullet (u^{\mathsf{T}} A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $u^{\mathsf{T}}\ f(A)\ v = 0$
- $f(A)\ v = 0$ means $u^{\mathsf{T}}\ f(A)\ v = 0$
- $\{f \mid f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}} = 0\} \subseteq \{f \mid f \bullet (u^{\mathsf{T}} A^i v)_{i \in \mathbb{Z}_{\geq 0}} = 0\}$
- Let $f_u^{A,v}$ be the minimal polynomial of $(u^{\mathsf{T}} A^i v)_{i \in \mathbb{Z}_{\geq 0}}$.
  - $f_u^{A,v} \mid f^{A,v} \mid f^A \mid \det(\lambda I - A)$
  - $\deg(f_u^{A,v}) \leq \deg(f^{A,v}) \leq \deg(f^A) \leq \deg(\det(\lambda I - A)) = n$

Computing the Minimal Polynomial of Krylov Sequence

Minimal Polynomials Probably Equal

If u chosen randomly from finite $S^n \subseteq \mathbb{F}^n$, then $f_u^{A,v} = f^{A,v}$ with probability at least

$$1 - \frac{\deg(f^{A,v})}{|S|}$$

Certificate

$$f_u^{A,v} \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}} = 0$$

Computing the Minimal Polynomial of Scalar Sequence

Degree Bound

Must know bound $M \geq \deg(f)$

Extended Euclidean Algorithm

- $s_j \; f_{-1} + t_j \; f_0 = f_j$
- Inputs are $f_{-1} = \lambda^{2M}$ and $f_0 = \sum_{i=0}^{2M-1} (a_i \; \lambda^i)$
- Stop when $\deg(f_j) \leq M < \deg(f_{j-1})$
- Minimal polynomial is reversal of $t_j(\lambda)$

Berlekamp-Massey Algorithm

- From coding theory
- Interpolates elements of scalar sequence
- Related to Extended Euclidean Algorithm

Wiedemann's Algorithm

Degree Bounds

- $f_u^{A,v} \mid f^{A,v} \mid f^A \mid \det(\lambda I - A)$
- $\deg(f_u^{A,v}) \leq \deg(\det(\lambda I - A)) = n$
- Only need $(u^\top A^i b)_{i=0}^{2n-1}$

Wiedemann's Algorithm

Require: nonsingular $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$

Ensure: $x \in \mathbb{F}^n$ such that $A\,x = b$

1: $u \leftarrow$ random vector in S^n where $S \subseteq \mathbb{F}$

2: use Berlekamp-Massey to compute
$f^{A,b} = c_0 + c_1\lambda + \cdots + c_m\lambda^m$ {Store only $A^i b$}

3: $x \leftarrow -\frac{1}{c_0}(c_1 b + c_2 Ab + \cdots c_m A^{m-1} b)$

Back to original problem

Example

- $$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 3 & 3 & -2 & 3 \\ 0 & -3 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix}$$
- $n = 4$
- $u^{\mathsf{T}} = \begin{bmatrix} 2 & 1 & 1 & 2 \end{bmatrix}$
- $(u^{\mathsf{T}} A^i b)_{i=0}^{2n-1} = (8, -4, 20, -28, 68, -124, 260, -508)$
- $f^{A,b} = f_u^{A,b} = \lambda^2 + \lambda - 2$
- $$x = \frac{1}{2}(Ab + b) = \frac{1}{2}\left(\begin{bmatrix} -2 \\ 2 \\ -6 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Extensions

Other Problems

- Randomly sample solution space of singular matrices
- Matrix rank
- Matrix determinant

Precondition Matrix

- $\tilde{A} = BA$ or $\tilde{A} = AB$
- Need efficient matrix-vector product
- New linear system $\tilde{A}\tilde{x} = \tilde{b}$